50. Make up two functions f_1 and f_2 that have the same Laplace transform. Do not think profound thoughts.

51. Reread *Remark* (*iii*) on page 224. Find the zero-input and the zero-state response for the IVP in Problem 38.

52. Suppose $f(t)$ is a function for which $f'(t)$ is piecewise continuous and of exponential order c. Use results in this section and Section 4.1 to justify

$$f(0) = \lim_{s\to\infty} sF(s),$$

where $F(s) = \mathscr{L}\{f(t)\}$. Verify this result with $f(t) = \cos kt$.

4.3 Translation Theorems

INTRODUCTION It is not convenient to use Definition 4.1.1 each time we wish to find the Laplace transform of a given function $f(t)$. For example, the integration by parts required to determine the transform of, say, $f(t) = e^t t^2 \sin 3t$ is formidable to say the least when done by hand. In this section and the next we present several labor-saving theorems that enable us to build up a more extensive list of transforms (see the table in Appendix III) without the necessity of using the definition of the Laplace transform.

4.3.1 Translation on the *s*-axis

Evaluating transforms such as $\mathscr{L}\{e^{5t}t^3\}$ and $\mathscr{L}\{e^{-2t}\cos 4t\}$ is straightforward provided we know $\mathscr{L}\{t^3\}$ and $\mathscr{L}\{\cos 4t\}$, which we do. In general, if we know $\mathscr{L}\{f(t)\} = F(s)$ it is possible to compute the Laplace transform of an exponential multiple of the function f, that is, $\mathscr{L}\{e^{at}f(t)\}$, with no additional effort other than *translating*, or *shifting*, $F(s)$ to $F(s - a)$. This result is known as the **first translation theorem** or **first shifting theorem**.

Theorem 4.3.1 First Translation Theorem

If $\mathscr{L}\{f(t)\} = F(s)$ and a is any real number, then

$$\mathscr{L}\{e^{at}f(t)\} = F(s - a).$$

PROOF: The proof is immediate, since by Definition 4.1.1

$$\mathscr{L}\{e^{at}f(t)\} = \int_0^\infty e^{-st}e^{at}f(t)\,dt = \int_0^\infty e^{-(s-a)t}f(t)\,dt = F(s - a). \qquad \equiv$$

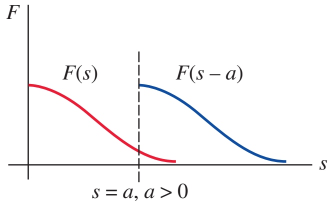


FIGURE 4.3.1 Shift on *s*-axis

If we consider s a real variable, then the graph of $F(s - a)$ is the graph of $F(s)$ shifted on the s-axis by the amount $|a|$. If $a > 0$, the graph of $F(s)$ is shifted a units to the right, whereas if $a < 0$, the graph is shifted $|a|$ units to the left. See **FIGURE 4.3.1**.

For emphasis it is sometimes useful to use the symbolism

$$\mathscr{L}\{e^{at}f(t)\} = \mathscr{L}\{f(t)\}_{s \to s-a},$$

where $s \to s - a$ means that in the Laplace transform $F(s)$ of $f(t)$ we replace the symbol s where it appears by $s - a$.

EXAMPLE 1 Using the First Translation Theorem

Evaluate **(a)** $\mathscr{L}\{e^{5t}t^3\}$ **(b)** $\mathscr{L}\{e^{-2t}\cos 4t\}$.

SOLUTION The results follow from Theorems 4.1.1 and 4.3.1.

(a) $$\mathscr{L}\{e^{5t}t^3\} = \mathscr{L}\{t^3\}_{s\to s-5} = \frac{3!}{s^4}\bigg|_{s\to s-5} = \frac{6}{(s-5)^4}$$

(b) $$\mathscr{L}\{e^{-2t}\cos 4t\} = \mathscr{L}\{\cos 4t\}_{s\to s-(-2)} = \frac{s}{s^2+16}\bigg|_{s\to s+2} = \frac{s+2}{(s+2)^2+16} \qquad \equiv$$

Inverse Form of Theorem 4.3.1 To compute the inverse of $F(s - a)$ we must recognize $F(s)$, find $f(t)$ by taking the inverse Laplace transform of $F(s)$, and then multiply $f(t)$ by the exponential function e^{at}. This procedure can be summarized symbolically in the following manner:

$$\mathscr{L}^{-1}\{F(s - a)\} = \mathscr{L}^{-1}\{F(s)|_{s\to s-a}\} = e^{at}f(t), \tag{1}$$

where $f(t) = \mathscr{L}^{-1}\{F(s)\}$.

EXAMPLE 2 Partial Fractions and Completing the Square

Partial fractions: repeated linear factors in (a).

Evaluate **(a)** $\mathscr{L}^{-1}\left\{\dfrac{2s+5}{(s-3)^2}\right\}$ **(b)** $\mathscr{L}^{-1}\left\{\dfrac{s/2+5/3}{s^2+4s+6}\right\}$.

SOLUTION **(a)** A repeated linear factor is a term $(s-a)^n$, where a is a real number and n is a positive integer ≥ 2. Recall that if $(s-a)^n$ appears in the denominator of a rational expression, then the assumed decomposition contains n partial fractions with constant numerators and denominators $s-a, (s-a)^2, \ldots, (s-a)^n$. Hence with $a = 3$ and $n = 2$ we write

$$\frac{2s+5}{(s-3)^2} = \frac{A}{s-3} + \frac{B}{(s-3)^2}.$$

The numerator obtained by putting the two terms on the right over a common denominator is $2s + 5 = A(s-3) + B$, and this identity yields $A = 2$ and $B = 11$. Therefore,

$$\frac{2s+5}{(s-3)^2} = \frac{2}{s-3} + \frac{11}{(s-3)^2} \tag{2}$$

and

$$\mathscr{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = 2\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} + 11\mathscr{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\}. \tag{3}$$

Now $1/(s-3)^2$ is $F(s) = 1/s^2$ shifted 3 units to the right. Since $\mathscr{L}^{-1}\{1/s^2\} = t$, it follows from (1) that

$$\mathscr{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} = \mathscr{L}^{-1}\left\{\frac{1}{s^2}\bigg|_{s\to s-3}\right\} = e^{3t}t.$$

Finally, (3) is

$$\mathscr{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = 2e^{3t} + 11e^{3t}t. \tag{4}$$

(b) To start, observe that the quadratic polynomial $s^2 + 4s + 6$ does not have real zeros and so has no real linear factors. In this situation we complete the square:

$$\frac{s/2+5/3}{s^2+4s+6} = \frac{s/2+5/3}{(s+2)^2+2}. \tag{5}$$

Our goal here is to recognize the expression on the right as some Laplace transform $F(s)$ in which s has been replaced throughout by $s + 2$. What we are trying to do here is analogous to working part (b) of Example 1 backward. The denominator in (5) is already in the correct form; that is, $s^2 + 2$ with s replaced by $s + 2$. However, we must fix up the numerator by manipulating the constants: $\frac{1}{2}s + \frac{5}{3} = \frac{1}{2}(s+2) + \frac{5}{3} - \frac{2}{2} = \frac{1}{2}(s+2) + \frac{2}{3}$.

Now by termwise division, the linearity of $\mathscr{L}^{-1}$, parts (d) and (e) of Theorem 4.2.1, and finally from (1),

$$\frac{s/2+5/3}{(s+2)^2+2} = \frac{(1/2)(s+2)+2/3}{(s+2)^2+2} = \frac{1}{2}\frac{s+2}{(s+2)^2+2} + \frac{2}{3}\frac{1}{(s+2)^2+2}$$

$$\begin{aligned}\mathscr{L}^{-1}\left\{\frac{s/2+5/3}{s^2+4s+6}\right\} &= \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\} + \frac{2}{3}\mathscr{L}^{-1}\left\{\frac{1}{(s+2)^2+2}\right\}\\ &= \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s}{s^2+2}\bigg|_{s\to s+2}\right\} + \frac{2}{3\sqrt{2}}\mathscr{L}^{-1}\left\{\frac{\sqrt{2}}{s^2+2}\bigg|_{s\to s+2}\right\} &(6)\\ &= \frac{1}{2}e^{-2t}\cos\sqrt{2}t + \frac{\sqrt{2}}{3}e^{-2t}\sin\sqrt{2}t. &(7)\end{aligned}$$

≡

EXAMPLE 3 Initial-Value Problem

Solve $y'' - 6y' + 9y = t^2e^{3t}$, $y(0) = 2$, $y'(0) = 17$.

SOLUTION Before transforming the DE note that its right-hand side is similar to the function in part (a) of Example 1. We use Theorem 4.3.1, the initial conditions, simplify, and then solve for $Y(s) = \mathscr{L}\{f(t)\}$:

$$\mathscr{L}\{y''\} - 6\mathscr{L}\{y'\} + 9\mathscr{L}\{y\} = \mathscr{L}\{t^2e^{3t}\}$$

$$s^2Y(s) - sy(0) - y'(0) - 6[sY(s) - y(0)] + 9Y(s) = \frac{2}{(s-3)^3}$$

$$(s^2 - 6s + 9)Y(s) = 2s + 5 + \frac{2}{(s-3)^3}$$

$$(s-3)^2Y(s) = 2s + 5 + \frac{2}{(s-3)^3}$$

$$Y(s) = \frac{2s+5}{(s-3)^2} + \frac{2}{(s-3)^5}.$$

The first term on the right has already been decomposed into individual partial fractions in (2) in part (a) of Example 2:

$$Y(s) = \frac{2}{s-3} + \frac{11}{(s-3)^2} + \frac{2}{(s-3)^5}.$$

Thus $\quad y(t) = 2\mathscr{L}^{-1}\left\{\frac{1}{s-3}\right\} + 11\mathscr{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} + \frac{2}{4!}\mathscr{L}^{-1}\left\{\frac{4!}{(s-3)^5}\right\}. \qquad (8)$

From the inverse form (1) of Theorem 4.3.1, the last two terms are

$$\mathscr{L}^{-1}\left\{\frac{1}{s^2}\bigg|_{s\to s-3}\right\} = te^{3t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{4!}{s^5}\bigg|_{s\to s-3}\right\} = t^4e^{3t},$$

and so (8) is $y(t) = 2e^{3t} + 11te^{3t} + \frac{1}{12}t^4e^{3t}$. ≡

EXAMPLE 4 An Initial-Value Problem

Solve $\quad y'' + 4y' + 6y = 1 + e^{-t}, \quad y(0) = 0, \quad y'(0) = 0.$

SOLUTION

$$\mathscr{L}\{y''\} + 4\mathscr{L}\{y'\} + 6\mathscr{L}\{y\} = \mathscr{L}\{1\} + \mathscr{L}\{e^{-t}\}$$

$$s^2Y(s) - sy(0) - y'(0) + 4[sY(s) - y(0)] + 6Y(s) = \frac{1}{s} + \frac{1}{s+1}$$

$$(s^2 + 4s + 6)Y(s) = \frac{2s+1}{s(s+1)}$$

$$Y(s) = \frac{2s+1}{s(s+1)(s^2+4s+6)}.$$

Since the quadratic term in the denominator does not factor into real linear factors, the partial fraction decomposition for $Y(s)$ is found to be

$$Y(s) = \frac{1/6}{s} + \frac{1/3}{s+1} - \frac{s/2 + 5/3}{s^2+4s+6}.$$

Moreover, in preparation for taking the inverse transform, we have already manipulated the last term into the necessary form in part (b) of Example 2. So in view of the results in (6) and (7) we have the solution

$$y(t) = \frac{1}{6}\mathscr{L}^{-1}\left\{\frac{1}{s}\right\} + \frac{1}{3}\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\} - \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\} - \frac{2}{3\sqrt{2}}\mathscr{L}^{-1}\left\{\frac{\sqrt{2}}{(s+2)^2+2}\right\}$$

$$= \frac{1}{6} + \frac{1}{3}e^{-t} - \frac{1}{2}e^{-2t}\cos\sqrt{2}t - \frac{\sqrt{2}}{3}e^{-2t}\sin\sqrt{2}t.$$ ≡

4.3.2 Translation on the *t*-axis

Unit Step Function In engineering, one frequently encounters functions that are either "off" or "on." For example, an external force acting on a mechanical system or a voltage impressed on a circuit can be turned off after a period of time. It is convenient, then, to define a special function that is the number 0 (off) up to a certain time $t = a$ and then the number 1 (on) after that time. This function is called the **unit step function** or the **Heaviside function**, named after the renowned English electrical engineer, physicist, and mathematician **Oliver Heaviside** (1850–1925). The Heaviside layer in the ionosphere, which can reflect radio waves, is named in his honor.

Definition 4.3.1 Unit Step Function

The **unit step function** $\mathcal{U}(t - a)$ is defined to be

$$\mathcal{U}(t-a) = \begin{cases} 0, & 0 \le t < a \\ 1, & t \ge a. \end{cases}$$

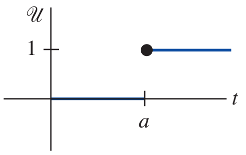


FIGURE 4.3.2 Graph of unit step function

Notice that we define $\mathcal{U}(t - a)$ only on the nonnegative t-axis since this is all that we are concerned with in the study of the Laplace transform. In a broader sense $\mathcal{U}(t - a) = 0$ for $t < a$. The graph of $\mathcal{U}(t - a)$ is given in **FIGURE 4.3.2**.

When a function f defined for $t \ge 0$ is multiplied by $\mathcal{U}(t - a)$, the unit step function "turns off" a portion of the graph of that function. For example, consider the function $f(t) = 2t - 3$. To "turn off" the portion of the graph of f on, say, the interval $0 \le t < 1$, we simply form the product $(2t - 3)\mathcal{U}(t - 1)$. See **FIGURE 4.3.3**. In general, the graph of $f(t)\mathcal{U}(t - a)$ is 0 (off) for $0 \le t < a$ and is the portion of the graph of f (on) for $t \ge a$.

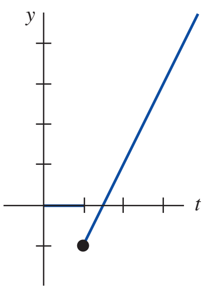


FIGURE 4.3.3 Function can be written $f(t) = (2t - 3)\mathcal{U}(t - 1)$

The unit step function can also be used to write piecewise-defined functions in a compact form. For example, by considering $0 \le t < 2$, $2 \le t < 3$, $t \ge 3$, and the corresponding values of $\mathcal{U}(t - 2)$ and $\mathcal{U}(t - 3)$, it should be apparent that the piecewise-defined function shown in **FIGURE 4.3.4** is the same as $f(t) = 2 - 3\,\mathcal{U}(t - 2) + \mathcal{U}(t - 3)$. Also, a general piecewise-defined function of the type

$$f(t) = \begin{cases} g(t), & 0 \le t < a \\ h(t), & t \ge a \end{cases} \tag{9}$$

is the same as

$$f(t) = g(t) - g(t)\,\mathcal{U}(t-a) + h(t)\,\mathcal{U}(t-a). \tag{10}$$

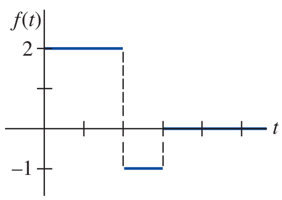


FIGURE 4.3.4 Function can be written $f(t) = 2 - 3\mathcal{U}(t - 2) + \mathcal{U}(t - 3)$

Similarly, a function of the type

$$f(t) = \begin{cases} 0, & 0 \le t < a \\ g(t), & a \le t < b \\ 0, & t \ge b \end{cases} \tag{11}$$

can be written
$$f(t) = g(t)[\mathcal{U}(t-a) - \mathcal{U}(t-b)]. \tag{12}$$

EXAMPLE 5 A Piecewise-Defined Function

Express $f(t) = \begin{cases} 20t, & 0 \le t < 5 \\ 0, & t \ge 5 \end{cases}$ in terms of unit step functions. Graph.

SOLUTION The graph of f is given in **FIGURE 4.3.5**. Now from (9) and (10) with $a = 5$, $g(t) = 20t$, and $h(t) = 0$, we get $f(t) = 20t - 20t\mathcal{U}(t - 5)$. ≡

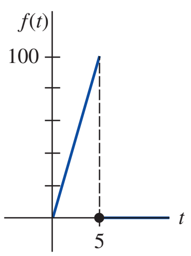
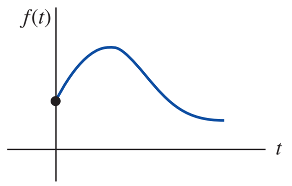


FIGURE 4.3.5 Function in Example 5

Consider a general function $y = f(t)$ defined for $t \ge 0$. The piecewise-defined function

$$f(t-a)\mathcal{U}(t-a) = \begin{cases} 0, & 0 \le t < a \\ f(t-a), & t \ge a \end{cases} \tag{13}$$



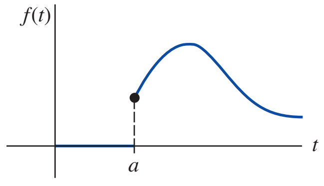


FIGURE 4.3.6 Shift on t-axis

plays a significant role in the discussion that follows. As shown in **FIGURE 4.3.6**, for $a > 0$ the graph of the function $y = f(t - a)\,\mathcal{U}(t - a)$ coincides with the graph of $y = f(t - a)$ for $t \geq a$ (which is the *entire* graph of $y = f(t)$, $t \geq 0$, shifted a units to the right on the t-axis) but is identically zero for $0 \leq t < a$.

We saw in Theorem 4.3.1 that an exponential multiple of $f(t)$ results in a translation of the transform $F(s)$ on the s-axis. As a consequence of the next theorem we see that whenever $F(s)$ is multiplied by an exponential function e^{-as}, $a > 0$, the inverse transform of the product $e^{-as}F(s)$ is the function f shifted along the t-axis in the manner illustrated in Figure 4.3.6(b). This result, presented next in its direct transform version, is called the **second translation theorem** or **second shifting theorem**.

Theorem 4.3.2 Second Translation Theorem

If $F(s) = \mathscr{L}\{f(t)\}$ and $a > 0$, then

$$\mathscr{L}\{f(t - a)\mathcal{U}(t - a)\} = e^{-as}F(s).$$

PROOF: By the additive interval property of integrals, $\int_0^\infty e^{-st}f(t - a)\,\mathcal{U}(t - a)\,dt$ can be written as two integrals:

$$\mathscr{L}\{f(t - a)\,\mathcal{U}(t - a)\} = \int_0^a e^{-st}f(t - a)\underbrace{\mathcal{U}(t - a)}_{\text{zero for } 0 \leq t < a}\,dt + \int_a^\infty e^{-st}f(t - a)\underbrace{\mathcal{U}(t - a)}_{\text{one for } t \geq a}\,dt$$

$$= \int_a^\infty e^{-st}f(t - a)\,dt.$$

Now if we let $v = t - a$, $dv = dt$ in the last integral, then

$$\mathscr{L}\{f(t - a)\,\mathcal{U}(t - a)\} = \int_0^\infty e^{-s(v + a)}f(v)\,dv$$

$$= e^{-as}\int_0^\infty e^{-sv}f(v)\,dv = e^{-as}\mathscr{L}\{f(t)\}. \qquad \equiv$$

We often wish to find the Laplace transform of just a unit step function. This can be found from either Definition 4.1.1 or Theorem 4.3.2. If we identify $f(t) = 1$ in Theorem 4.3.2, then $f(t - a) = 1$, $F(s) = \mathscr{L}\{1\} = 1/s$, and so

$$\mathscr{L}\{\mathcal{U}(t - a)\} = \frac{e^{-as}}{s}. \tag{14}$$

EXAMPLE 6 Figure 4.3.4 Revisited

Find the Laplace transform of the function f whose graph is given in Figure 4.3.4.

SOLUTION We use f expressed in terms of the unit step function

$$f(t) = 2 - 3\,\mathcal{U}(t - 2) + \mathcal{U}(t - 3)$$

and the result given in (14):

$$\mathscr{L}\{f(t)\} = 2\mathscr{L}\{1\} - 3\mathscr{L}\{\mathcal{U}(t - 2)\} + \mathscr{L}\{\mathcal{U}(t - 3)\}$$

$$= \frac{2}{s} - \frac{3e^{-2s}}{s} + \frac{e^{-3s}}{s}. \qquad \equiv$$

Inverse Form of Theorem 4.3.2 If $f(t) = \mathscr{L}^{-1}\{F(s)\}$, the inverse form of Theorem 4.3.2, with $a > 0$, is

$$\mathscr{L}^{-1}\{e^{-as}\,F(s)\} = f(t - a)\mathcal{U}(t - a). \tag{15}$$

EXAMPLE 7 Using Formula (15)

Evaluate **(a)** $\mathscr{L}^{-1}\left\{\dfrac{1}{s-4}e^{-2s}\right\}$ **(b)** $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2+9}e^{-\pi s/2}\right\}$.

SOLUTION **(a)** With the identifications $a = 2$, $F(s) = 1/(s-4)$, $\mathscr{L}^{-1}\{F(s)\} = e^{4t}$, we have from (15)

$$\mathscr{L}^{-1}\left\{\frac{1}{s-4}e^{-2s}\right\} = e^{4(t-2)}\mathscr{U}(t-2).$$

(b) With $a = \pi/2$, $F(s) = s/(s^2+9)$, $\mathscr{L}^{-1}\{F(s)\} = \cos 3t$, (15) yields

$$\mathscr{L}^{-1}\left\{\frac{s}{s^2+9}e^{-\pi s/2}\right\} = \cos 3(t-\pi/2)\,\mathscr{U}(t-\pi/2).$$

The last expression can be simplified somewhat using the addition formula for the cosine. Verify that the result is the same as $-\sin 3t\,\mathscr{U}(t-\pi/2)$. ≡

Alternative Form of Theorem 4.3.2

We are frequently confronted with the problem of finding the Laplace transform of a product of a function g and a unit step function $\mathscr{U}(t-a)$ where the function g lacks the precise shifted form $f(t-a)$ in Theorem 4.3.2. To find the Laplace transform of $g(t)\mathscr{U}(t-a)$, it is possible to fix up $g(t)$ into the required form $f(t-a)$ by algebraic manipulations. For example, if we want to use Theorem 4.3.2 to find the Laplace transform of $t^2\,\mathscr{U}(t-2)$, we would have to force $g(t) = t^2$ into the form $f(t-2)$. You should work through the details and verify that $t^2 = (t-2)^2 + 4(t-2) + 4$ is an identity. Therefore,

$$\begin{aligned}\mathscr{L}\{t^2\,\mathscr{U}(t-2)\} &= \mathscr{L}\{(t-2)^2\,\mathscr{U}(t-2) + 4(t-2)\,\mathscr{U}(t-2) + 4\,\mathscr{U}(t-2)\},\\ &= \frac{2e^{-2s}}{s^3} + \frac{4e^{-2s}}{s^2} + \frac{4e^{-2s}}{s}\end{aligned}$$

by Theorem 4.3.2. But since these manipulations are time-consuming and often not obvious, it is simpler to devise an alternative version of Theorem 4.3.2. Using Definition 4.1.1, the definition of $\mathscr{U}(t-a)$, and the substitution $u = t-a$, we obtain

$$\mathscr{L}\{g(t)\mathscr{U}(t-a)\} = \int_a^\infty e^{-st}g(t)\,dt = \int_0^\infty e^{-s(u+a)}g(u+a)\,du.$$

That is, $$\mathscr{L}\{g(t)\mathscr{U}(t-a)\} = e^{-as}\mathscr{L}\{g(t+a)\}. \tag{16}$$

With $a = 2$ and $g(t) = t^2$ we have by (16),

$$\begin{aligned}\mathscr{L}\{t^2\mathscr{U}(t-2)\} &= e^{-2s}\mathscr{L}\{(t+2)^2\}\\ &= e^{-2s}\mathscr{L}\{t^2+4t+4\}\\ &= e^{-2s}\left(\frac{2}{s^3}+\frac{4}{s^2}+\frac{4}{s}\right),\end{aligned}$$

which agrees with our previous calculation.

EXAMPLE 8 Second Translation Theorem—Alternative Form

Evaluate $\mathscr{L}\{\cos t\,\mathscr{U}(t-\pi)\}$.

SOLUTION With $g(t) = \cos t$, $a = \pi$, then $g(t+\pi) = \cos(t+\pi) = -\cos t$ by the addition formula for the cosine function. Hence by (16),

$$\mathscr{L}\{\cos t\,\mathscr{U}(t-\pi)\} = -e^{-\pi s}\mathscr{L}\{\cos t\} = -\frac{s}{s^2+1}e^{-\pi s}.$$ ≡

In the next two examples, we solve in turn an initial-value problem and a boundary-value problem. Both problems involve a piecewise-linear differential equation.

EXAMPLE 9 An Initial-Value Problem

Solve $y' + y = f(t), \quad y(0) = 5, \quad$ where $\quad f(t) = \begin{cases} 0, & 0 \le t < \pi \\ 3\cos t, & t \ge \pi. \end{cases}$

SOLUTION The function f can be written as $f(t) = 3\cos t\,\mathcal{U}(t - \pi)$ and so by linearity, the results of Example 8, and the usual partial fractions, we have

$$\mathscr{L}\{y'\} + \mathscr{L}\{y\} = 3\mathscr{L}\{\cos t\,\mathcal{U}(t-\pi)\}$$

$$sY(s) - y(0) + Y(s) = -3\frac{s}{s^2+1}e^{-\pi s}$$

$$(s+1)Y(s) = 5 - \frac{3s}{s^2+1}e^{-\pi s}$$

$$Y(s) = \frac{5}{s+1} - \frac{3}{2}\left[-\frac{1}{s+1}e^{-\pi s} + \frac{1}{s^2+1}e^{-\pi s} + \frac{s}{s^2+1}e^{-\pi s}\right]. \tag{17}$$

Now proceeding as we did in Example 7, it follows from (15) with $a = \pi$ that the inverses of the terms in the bracket are

$$\mathscr{L}^{-1}\left\{\frac{1}{s+1}e^{-\pi s}\right\} = e^{-(t-\pi)}\mathcal{U}(t-\pi), \quad \mathscr{L}^{-1}\left\{\frac{1}{s^2+1}e^{-\pi s}\right\} = \sin(t-\pi)\,\mathcal{U}(t-\pi),$$

and

$$\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}e^{-\pi s}\right\} = \cos(t-\pi)\,\mathcal{U}(t-\pi).$$

Thus the inverse of (17) is

$$y(t) = 5e^{-t} + \frac{3}{2}e^{-(t-\pi)}\mathcal{U}(t-\pi) - \frac{3}{2}\sin(t-\pi)\,\mathcal{U}(t-\pi) - \frac{3}{2}\cos(t-\pi)\,\mathcal{U}(t-\pi)$$

$$= 5e^{-t} + \frac{3}{2}\left[e^{-(t-\pi)} + \sin t + \cos t\right]\mathcal{U}(t-\pi) \quad \leftarrow \text{trigonometric identities}$$

$$= \begin{cases} 5e^{-t}, & 0 \le t < \pi \\ 5e^{-t} + \frac{3}{2}e^{-(t-\pi)} + \frac{3}{2}\sin t + \frac{3}{2}\cos t, & t \ge \pi. \end{cases} \tag{18}$$

With the aid of a graphing utility we get the graph of (18), shown in **FIGURE 4.3.7**. ≡

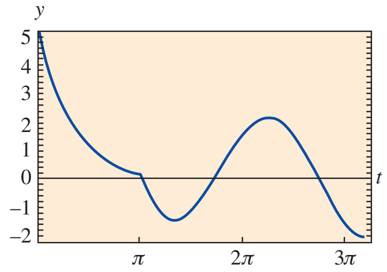


FIGURE 4.3.7 Graph of function (18) in Example 9

Beams In Section 3.9 we saw that the static deflection $y(x)$ of a uniform beam of length L carrying load $w(x)$ per unit length is found from the linear fourth-order differential equation

$$EI\frac{d^4y}{dx^4} = w(x), \tag{19}$$

where E is Young's modulus of elasticity and I is a moment of inertia of a cross section of the beam. The Laplace transform is particularly useful when $w(x)$ is piecewise defined, but in order to use the transform, we must tacitly assume that $y(x)$ and $w(x)$ are defined on $(0, \infty)$ rather than on $(0, L)$. Note, too, that the next example is a boundary-value problem rather than an initial-value problem.

EXAMPLE 10 A Boundary-Value Problem

A beam of length L is embedded at both ends as shown in **FIGURE 4.3.8**. Find the deflection of the beam when the load is given by

$$w(x) = \begin{cases} w_0\left(1 - \frac{2}{L}x\right), & 0 < x \le L/2 \\ 0, & L/2 < x < L, \end{cases}$$

where w_0 is a constant.

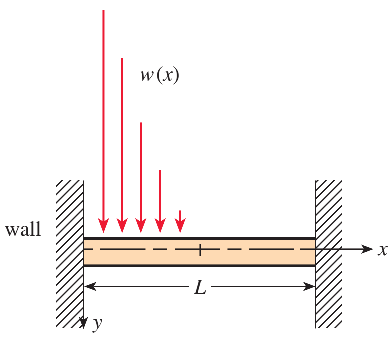


FIGURE 4.3.8 Embedded beam with a variable load in Example 10

SOLUTION Recall that, since the beam is embedded at both ends, the boundary conditions are $y(0) = 0$, $y'(0) = 0$, $y(L) = 0$, $y'(L) = 0$. Now by (10) we can express $w(x)$ in terms of the unit step function:

$$\begin{aligned} w(x) &= w_0\left(1 - \frac{2}{L}x\right) - w_0\left(1 - \frac{2}{L}x\right)\mathscr{U}\left(x - \frac{L}{2}\right) \\ &= \frac{2w_0}{L}\left[\frac{L}{2} - x + \left(x - \frac{L}{2}\right)\mathscr{U}\left(x - \frac{L}{2}\right)\right]. \end{aligned}$$

Transforming (19) with respect to the variable x gives

$$EI(s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)) = \frac{2w_0}{L}\left[\frac{L/2}{s} - \frac{1}{s^2} + \frac{1}{s^2}e^{-Ls/2}\right]$$

or

$$s^4Y(s) - sy''(0) - y'''(0) = \frac{2w_0}{EIL}\left[\frac{L/2}{s} - \frac{1}{s^2} + \frac{1}{s^2}e^{-Ls/2}\right].$$

If we let $c_1 = y''(0)$ and $c_2 = y'''(0)$, then

$$Y(s) = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{2w_0}{EIL}\left[\frac{L/2}{s^5} - \frac{1}{s^6} + \frac{1}{s^6}e^{-Ls/2}\right],$$

and consequently

$$\begin{aligned} y(x) &= \frac{c_1}{2!}\mathscr{L}^{-1}\left\{\frac{2!}{s^3}\right\} + \frac{c_2}{3!}\mathscr{L}^{-1}\left\{\frac{3!}{s^4}\right\} \\ &\quad + \frac{2w_0}{EIL}\left[\frac{L/2}{4!}\mathscr{L}^{-1}\left\{\frac{4!}{s^5}\right\} - \frac{1}{5!}\mathscr{L}^{-1}\left\{\frac{5!}{s^6}\right\} + \frac{1}{5!}\mathscr{L}^{-1}\left\{\frac{5!}{s^6}e^{-Ls/2}\right\}\right] \\ &= \frac{c_1}{2}x^2 + \frac{c_2}{6}x^3 + \frac{w_0}{60EIL}\left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2}\right)^5\mathscr{U}\left(x - \frac{L}{2}\right)\right]. \end{aligned}$$

Applying the conditions $y(L) = 0$ and $y'(L) = 0$ to the last result yields a system of equations for c_1 and c_2:

$$\begin{aligned} c_1\frac{L^2}{2} + c_2\frac{L^3}{6} + \frac{49w_0L^4}{1920EI} &= 0 \\ c_1L + c_2\frac{L^2}{2} + \frac{85w_0L^3}{960EI} &= 0. \end{aligned}$$

Solving, we find $c_1 = 23w_0L^2/960EI$ and $c_2 = -9w_0L/40EI$. Thus the deflection is

$$y(x) = \frac{23w_0L^2}{1920EI}x^2 - \frac{3w_0L}{80EI}x^3 + \frac{w_0}{60EIL}\left[\frac{5L}{2}x^4 - x^5 + \left(x - \frac{L}{2}\right)^5\mathscr{U}\left(x - \frac{L}{2}\right)\right]. \quad \equiv$$

REMARKS

Outside the discussion of the Laplace transform, the unit step function is defined on the interval $(-\infty, \infty)$, that is,

$$\mathscr{U}(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a. \end{cases}$$

Using this slight modification of Definition 4.3.1, a special case of (12) when $g(t) = 1$ is sometimes called the **boxcar function** and denoted by

$$\Pi(t) = \mathscr{U}(t - a) - \mathscr{U}(t - b).$$

See **FIGURE 4.3.9**.

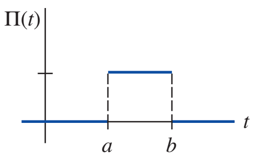


FIGURE 4.3.9 Boxcar function

4.3 Exercises

Answers to selected odd-numbered problems begin on page ANS-9.

4.3.1 Translation on the *s*-axis

In Problems 1–20, find either $F(s)$ or $f(t)$, as indicated.

1. $\mathscr{L}\{te^{10t}\}$

2. $\mathscr{L}\{te^{-6t}\}$

3. $\mathscr{L}\{t^3e^{-2t}\}$

4. $\mathscr{L}\{t^{10}e^{-7t}\}$

5. $\mathscr{L}\{t(e^t + e^{2t})^2\}$

6. $\mathscr{L}\{e^{2t}(t-1)^2\}$

7. $\mathscr{L}\{e^t \sin 3t\}$

8. $\mathscr{L}\{e^{-2t} \cos 4t\}$

9. $\mathscr{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\}$

10. $\mathscr{L}\left\{e^{3t}\left(9 - 4t + 10 \sin \dfrac{t}{2}\right)\right\}$

11. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s+2)^3}\right\}$

12. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)^4}\right\}$

13. $\mathscr{L}^{-1}\left\{\dfrac{1}{s^2 - 6s + 10}\right\}$

14. $\mathscr{L}^{-1}\left\{\dfrac{1}{s^2 + 2s + 5}\right\}$

15. $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2 + 4s + 5}\right\}$

16. $\mathscr{L}^{-1}\left\{\dfrac{2s + 5}{s^2 + 6s + 34}\right\}$

17. $\mathscr{L}^{-1}\left\{\dfrac{s}{(s+1)^2}\right\}$

18. $\mathscr{L}^{-1}\left\{\dfrac{5s}{(s-2)^2}\right\}$

19. $\mathscr{L}^{-1}\left\{\dfrac{2s - 1}{s^2(s+1)^3}\right\}$

20. $\mathscr{L}^{-1}\left\{\dfrac{(s+1)^2}{(s+2)^4}\right\}$

In Problems 21–30, use the Laplace transform to solve the given initial-value problem.

21. $y' + 4y = e^{-4t}, \quad y(0) = 2$

22. $y' - y = 1 + te^t, \quad y(0) = 0$

23. $y'' + 2y' + y = 0, \quad y(0) = 1, \quad y'(0) = 1$

24. $y'' - 4y' + 4y = t^3e^{2t}, \quad y(0) = 0, \quad y'(0) = 0$

25. $y'' - 6y' + 9y = t, \quad y(0) = 0, \quad y'(0) = 1$

26. $y'' - 4y' + 4y = t^3, \quad y(0) = 1, \quad y'(0) = 0$

27. $y'' - 6y' + 13y = 0, \quad y(0) = 0, \quad y'(0) = -3$

28. $2y'' + 20y' + 51y = 0, \quad y(0) = 2, \quad y'(0) = 0$

29. $y'' - y' = e^t \cos t, \quad y(0) = 0, \quad y'(0) = 0$

30. $y'' - 2y' + 5y = 1 + t, \quad y(0) = 0, \quad y'(0) = 4$

In Problems 31 and 32, use the Laplace transform and the procedure outlined in Example 10 to solve the given boundary-value problem.

31. $y'' + 2y' + y = 0, \quad y'(0) = 2, \quad y(1) = 2$

32. $y'' + 8y' + 20y = 0, \quad y(0) = 0, \quad y'(\pi) = 0$

33. A 4-lb weight stretches a spring 2 ft. The weight is released from rest 18 in above the equilibrium position, and the resulting motion takes place in a medium offering a damping force numerically equal to $\frac{7}{8}$ times the instantaneous velocity. Use the Laplace transform to find the equation of motion $x(t)$.

34. Recall that the differential equation for the instantaneous charge $q(t)$ on the capacitor in an *LRC*-series circuit is

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t). \tag{20}$$

See Section 3.8. Use the Laplace transform to find $q(t)$ when $L = 1$ h, $R = 20\ \Omega$, $C = 0.005$ f, $E(t) = 150$ V, $t > 0$, $q(0) = 0$, and $i(0) = 0$. What is the current $i(t)$?

35. Consider the battery of constant voltage E_0 that charges the capacitor shown in **FIGURE 4.3.10**. Divide equation (20) by L and define $2\lambda = R/L$ and $\omega^2 = 1/LC$. Use the Laplace transform to show that the solution $q(t)$ of $q'' + 2\lambda q' + \omega^2 q = E_0/L$, subject to $q(0) = 0$, $i(0) = 0$, is

$$q(t) = \begin{cases} E_0C\left[1 - e^{-\lambda t}\left(\cosh\sqrt{\lambda^2 - \omega^2}\,t + \dfrac{\lambda}{\sqrt{\lambda^2 - \omega^2}}\sinh\sqrt{\lambda^2 - \omega^2}\,t\right)\right], & \lambda > \omega \\ E_0C\left[1 - e^{-\lambda t}(1 + \lambda t)\right], & \lambda = \omega \\ E_0C\left[1 - e^{-\lambda t}\left(\cos\sqrt{\omega^2 - \lambda^2}\,t + \dfrac{\lambda}{\sqrt{\omega^2 - \lambda^2}}\sin\sqrt{\omega^2 - \lambda^2}\,t\right)\right], & \lambda < \omega. \end{cases}$$

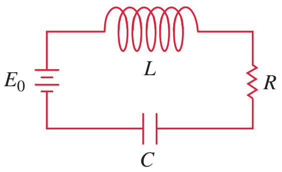


FIGURE 4.3.10 Circuit in Problem 35

36. Use the Laplace transform to find the charge $q(t)$ in an *RC*-series when $q(0) = 0$ and $E(t) = E_0e^{-kt}$, $k > 0$. Consider two cases: $k \neq 1/RC$ and $k = 1/RC$.

4.3.2 Translation on the *t*-axis

In Problems 37–48, find either $F(s)$ or $f(t)$, as indicated.

37. $\mathscr{L}\{(t-1)\,\mathscr{U}(t-1)\}$

38. $\mathscr{L}\{e^{2-t}\,\mathscr{U}(t-2)\}$

39. $\mathscr{L}\{t\,\mathscr{U}(t-2)\}$

40. $\mathscr{L}\{(3t+1)\,\mathscr{U}(t-1)\}$

41. $\mathscr{L}\{\cos 2t\,\mathscr{U}(t-\pi)\}$

42. $\mathscr{L}\{\sin t\,\mathscr{U}(t - \pi/2)\}$

43. $\mathscr{L}^{-1}\left\{\dfrac{e^{-2s}}{s^3}\right\}$

44. $\mathscr{L}^{-1}\left\{\dfrac{(1 + e^{-2s})^2}{s + 2}\right\}$

45. $\mathscr{L}^{-1}\left\{\dfrac{e^{-\pi s}}{s^2 + 1}\right\}$

46. $\mathscr{L}^{-1}\left\{\dfrac{se^{-\pi s/2}}{s^2 + 4}\right\}$

47. $\mathscr{L}^{-1}\left\{\dfrac{e^{-s}}{s(s+1)}\right\}$

48. $\mathscr{L}^{-1}\left\{\dfrac{e^{-2s}}{s^2(s-1)}\right\}$

In Problems 49–54, match the given graph with one of the given functions in (a)–(f). The graph of $f(t)$ is given in **FIGURE 4.3.11**.

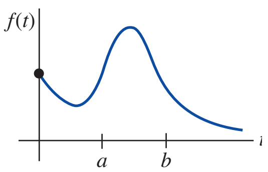


FIGURE 4.3.11 Graph for Problems 49–54

(a) $f(t) - f(t)\,\mathcal{U}(t-a)$
(b) $f(t-b)\,\mathcal{U}(t-b)$
(c) $f(t)\,\mathcal{U}(t-a)$
(d) $f(t) - f(t)\,\mathcal{U}(t-b)$
(e) $f(t)\,\mathcal{U}(t-a) - f(t)\,\mathcal{U}(t-b)$
(f) $f(t-a)\,\mathcal{U}(t-a) - f(t-a)\,\mathcal{U}(t-b)$

49.

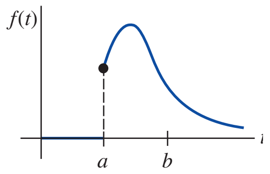


FIGURE 4.3.12 Graph for Problem 49

50.

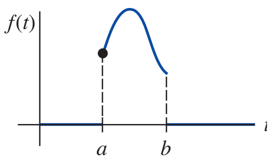


FIGURE 4.3.13 Graph for Problem 50

51.

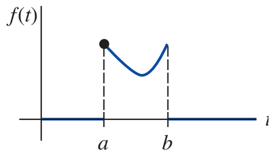


FIGURE 4.3.14 Graph for Problem 51

52.

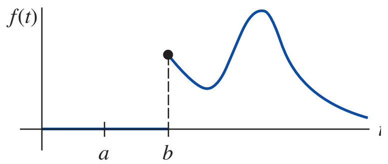


FIGURE 4.3.15 Graph for Problem 52

53.

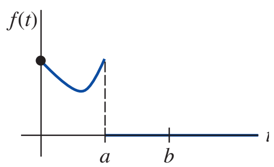


FIGURE 4.3.16 Graph for Problem 53

54.

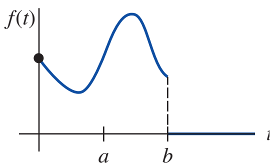


FIGURE 4.3.17 Graph for Problem 54

In Problems 55–62, write each function in terms of unit step functions. Find the Laplace transform of the given function.

55. $f(t) = \begin{cases} 2, & 0 \le t < 3 \\ -2, & t \ge 3 \end{cases}$

56. $f(t) = \begin{cases} 1, & 0 \le t < 4 \\ 0, & 4 \le t < 5 \\ 1, & t \ge 5 \end{cases}$

57. $f(t) = \begin{cases} 0, & 0 \le t < 1 \\ t^2, & t \ge 1 \end{cases}$

58. $f(t) = \begin{cases} 0, & 0 \le t < 3\pi/2 \\ \sin t, & t \ge 3\pi/2 \end{cases}$

59. $f(t) = \begin{cases} t, & 0 \le t < 2 \\ 0, & t \ge 2 \end{cases}$

60. $f(t) = \begin{cases} \sin t, & 0 \le t < 2\pi \\ 0, & t \ge 2\pi \end{cases}$

61.

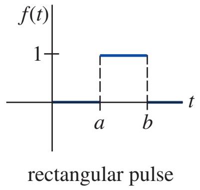


FIGURE 4.3.18 Graph for Problem 61

62.

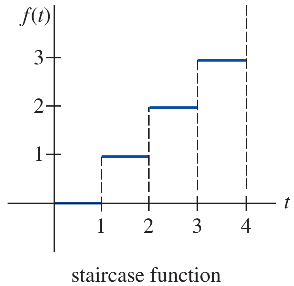


FIGURE 4.3.19 Graph for Problem 62

In Problems 63–70, use the Laplace transform to solve the given initial-value problem.

63. $y' + y = f(t), \quad y(0) = 0, \text{ where } f(t) = \begin{cases} 0, & 0 \le t < 1 \\ 5, & t \ge 1 \end{cases}$

64. $y' + y = f(t), \quad y(0) = 0, \text{ where } f(t) = \begin{cases} 1, & 0 \le t < 1 \\ -1, & t \ge 1 \end{cases}$

65. $y' + 2y = f(t), \quad y(0) = 0, \text{ where } f(t) = \begin{cases} t, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$

66. $y'' + 4y = f(t), \quad y(0) = 0, \; y'(0) = -1$, where

$$f(t) = \begin{cases} 1, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$$

67. $y'' + 4y = \sin t\,\mathcal{U}(t - 2\pi), \quad y(0) = 1, \; y'(0) = 0$

68. $y'' - 5y' + 6y = \mathcal{U}(t-1), \quad y(0) = 0, \; y'(0) = 1$

69. $y'' + y = f(t), \quad y(0) = 0, \; y'(0) = 1$, where

$$f(t) = \begin{cases} 0, & 0 \le t < \pi \\ 1, & \pi \le t < 2\pi \\ 0, & t \ge 2\pi \end{cases}$$

70. $y'' + 4y' + 3y = 1 - \mathcal{U}(t-2) - \mathcal{U}(t-4) + \mathcal{U}(t-6)$, $y(0) = 0, \; y'(0) = 0$

71. Suppose a mass weighing 32 lb stretches a spring 2 ft. If the weight is released from rest at the equilibrium position, find the equation of motion $x(t)$ if an impressed force $f(t) = 20t$ acts on the system for $0 \le t < 5$ and is then removed (see Example 5). Ignore any damping forces. Use a graphing utility to obtain the graph $x(t)$ on the interval $[0, 10]$.

72. Solve Problem 71 if the impressed force $f(t) = \sin t$ acts on the system for $0 \le t < 2\pi$ and is then removed.

In Problems 73 and 74, use the Laplace transform to find the charge $q(t)$ on the capacitor in an RC-series circuit subject to the given conditions.

73. $q(0) = 0, R = 2.5\ \Omega$, $C = 0.08$ f, $E(t)$ given in **FIGURE 4.3.20**

74. $q(0) = q_0, R = 10\ \Omega, C = 0.1$ f, $E(t)$ given in **FIGURE 4.3.21**